Vidyavardhini's College of Engineering and Technology

Department of Artificial Intelligence & Data Science

AY: 2026-27

Class:	**TE-AI&DS**	**Semester:**	**V**
Course Code:	**2015111**	**Course Name:**	**Statistics for Machine Learning and Data Science Lab**

Name of Student:	**ASHLESHA .S. BHOIR**
Roll No. :	**03**
Experiment No.:	**4**
Title of the Experiment:	**Perform Statistical Inference, Estimation, and Hypothesis Testing on Real-World Datasets**
Date of Performance:	
Date of Submission:	

Evaluation

Performance Indicator	Max. Marks	Marks Obtained
Performance	5	
Understanding	5	
Journal work and timely submission	10	
Total	20	

Performance Indicator	Exceed Expectations (EE)	Meet Expectations (ME)	Below Expectations (BE)
Performance	4-5	2-3	1
Understanding	4-5	2-3	1
Journal work and timely submission	8-10	5-8	1-4

Checked by

Name of Faculty :

Signature :

Date :

Aim: To apply statistical estimation and hypothesis testing techniques to the Pima Indians Diabetes Dataset and draw conclusions about population characteristics using sample data.

Objective

After completing this experiment, the student will be able to:

1. Apply estimation and hypothesis testing techniques to analyze population characteristics using sample data.
2. Interpret statistical test results and make data-driven conclusions based on the obtained evidence.

Tools Required

1. Python 3.x
2. Google Colab / Jupyter Notebook
3. Pandas
4. NumPy
5. SciPy
6. Matplotlib
7. Seaborn

Dataset Details

Dataset: Pima Indians Diabetes Dataset

The dataset contains diagnostic information for 768 female patients

Procedure

1. Load the Pima Indians Diabetes Dataset.
2. Select appropriate variables and divide the data into suitable groups where required.
3. Calculate point estimates such as sample mean and sample proportion.
4. Construct confidence intervals for selected population parameters.

5. Formulate null and alternative hypotheses for selected research questions.
6. Select and apply an appropriate statistical test, such as:
    a. One-sample t-test
    b. Two-sample t-test
    c. Chi-square test
7. Calculate the test statistic and p-value.
8. Compare the p-value with the selected significance level.
9. Make a statistical decision and interpret the result in the context of the dataset.

Description of the Experiment

Statistical inference allows conclusions about a population to be drawn using information from a sample.

The general process is: Sample Data, Estimate Population Parameter, Construct Confidence Interval, Formulate Hypotheses, Perform Statistical Test, and Interpret Results

For example, a sample of patients can be used to investigate whether the average glucose level differs from a specified reference value or whether diabetes outcome is associated with another categorical variable.

Detailed Description of Statistical Methods

Point Estimation

A sample statistic is used to estimate an unknown population parameter.

For example:

$$x^{-}$$

is used to estimate the population mean μ.

A sample proportion is:

$$\hat{p} = \frac{x}{n}$$

where $x$is the number of observations with a particular characteristic.

Confidence Interval

A confidence interval provides a range of plausible values for a population parameter.

For a mean:

$$\bar{x} \pm t_{\alpha/2}\frac{s}{\sqrt{n}}$$

The confidence interval should be interpreted in the context of the population parameter being estimated.

Hypothesis Testing

A hypothesis test evaluates a claim about a population.

Null Hypothesis

$$H_0$$

represents the initial assumption.

Alternative Hypothesis

$$H_1$$

represents the claim being investigated.

A significance level such as:

$$\alpha = 0.05$$

may be used.

Decision Rule

- If $p - value < \alpha$, reject H_0.
- If $p - value \geq \alpha$, fail to reject H_0.

t-Test

A t-test can be used to compare means.

Examples:

- Test whether the mean glucose level differs from a specified value.
- Compare the mean BMI of two groups.

Chi-Square Test

The chi-square test can be used to test the association between categorical variables.

For example:

$$Diabetes\ Outcome and Gender$$

where applicable.

Implementation

```
[1]
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from scipy import stats
```

```
[2]
url = "https://raw.githubusercontent.com/jbrownlee/Datasets/master/pima-indians-diabetes.data.csv"

columns = [
    "Pregnancies",
    "Glucose",
    "BloodPressure",
    "SkinThickness",
    "Insulin",
    "BMI",
    "DiabetesPedigreeFunction",
    "Age",
    "Outcome"
]
```

```
[2]
]

df = pd.read_csv(url, names=columns)

print("Dataset loaded successfully!")
print("Shape of dataset:", df.shape)

df.head()
```

```
Dataset loaded successfully!
Shape of dataset: (768, 9)
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

[3]
✓ 0s

```
print("Dataset Information:")
print(df.info())

print("\nMissing Values:")
print(df.isnull().sum())

print("\nStatistical Summary:")
df.describe()
```

```
Dataset Information:
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
```

```
Missing Values:
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64

Statistical Summary:
```

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	0.471876	33.240885	0.348958
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	21.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.243750	24.000000	0.000000

[4]
0s

```
glucose = df["Glucose"]

sample_mean = glucose.mean()
sample_std = glucose.std()
sample_size = len(glucose)

print("Point Estimation of Population Mean Glucose Level")
print("---------------------------------------------")
print("Sample Size (n):", sample_size)
print("Sample Mean (x̄):", round(sample_mean, 2))
print("Sample Standard Deviation (s):", round(sample_std, 2))
```

```
Point Estimation of Population Mean Glucose Level
---------------------------------------------
Sample Size (n): 768
Sample Mean (x̄): 120.89
Sample Standard Deviation (s): 31.97
```

[5]
0s

```
confidence_level = 0.95
alpha = 1 - confidence_level

t_critical = stats.t.ppf(1 - alpha/2, df=sample_size - 1)

margin_of_error = t_critical * (sample_std / np.sqrt(sample_size))

lower_bound = sample_mean - margin_of_error
upper_bound = sample_mean + margin_of_error

print("95% Confidence Interval for Mean Glucose Level")
print("-----------------------------------------------")
print("Sample Mean:", round(sample_mean, 2))
print("Margin of Error:", round(margin_of_error, 2))
print("Lower Bound:", round(lower_bound, 2))
print("Upper Bound:", round(upper_bound, 2))
print(
    f"95% Confidence Interval: ({lower_bound:.2f}, {upper_bound:.2f})"
)
```

```
95% Confidence Interval for Mean Glucose Level
-----------------------------------------------
Sample Mean: 120.89
Margin of Error: 2.26
Lower Bound: 118.63
Upper Bound: 123.16
95% Confidence Interval: (118.63, 123.16)
```

[6]
0s

```python
reference_mean = 120

t_statistic, p_value = stats.ttest_1samp(glucose, reference_mean)

print("One-Sample t-Test")
print("-----------------")
print("H0: μ = 120 mg/dL")
print("H1: μ ≠ 120 mg/dL")
print("Significance Level (α): 0.05")
print()
print("Test Statistic (t):", round(t_statistic, 4))
print("P-value:", round(p_value, 4))

if p_value < 0.05:
```

[6]
0s

```python
if p_value < 0.05:
    print("Decision: Reject H0")
    print("Conclusion: There is significant evidence that the mean glucose level differs from 120 mg/dL.")
else:
    print("Decision: Fail to Reject H0")
    print("Conclusion: There is insufficient evidence that the mean glucose level differs from 120 mg/dL.")
```

```
One-Sample t-Test
-----------------
H0: μ = 120 mg/dL
H1: μ ≠ 120 mg/dL
Significance Level (α): 0.05

Test Statistic (t): 0.7754
P-value: 0.4384
Decision: Fail to Reject H0
Conclusion: There is insufficient evidence that the mean glucose level differs from 120 mg/dL.
```

```
[7]
0s
bmi_non_diabetic = df[df["Outcome"] == 0]["BMI"]
bmi_diabetic = df[df["Outcome"] == 1]["BMI"]

t_statistic, p_value = stats.ttest_ind(
    bmi_diabetic,
    bmi_non_diabetic,
    equal_var=False
)

print("Two-Sample t-Test: BMI")
print("----------------------")
print("H0: Mean BMI of diabetic and non-diabetic patients is equal")
print("H1: Mean BMI of diabetic and non-diabetic patients is different")
print()
print("Mean BMI - Non-Diabetic:", round(bmi_non_diabetic.mean(), 2))
print("Mean BMI - Diabetic:", round(bmi_diabetic.mean(), 2))
print("Test Statistic (t):", round(t_statistic, 4))
print("P-value:", round(p_value, 4))

if p_value < 0.05:
    print("Decision: Reject H0")
```


```
if p_value < 0.05:
    print("Decision: Reject H0")
```


```
    print("Conclusion: There is a significant difference in mean BMI between the two groups.")
else:
    print("Decision: Fail to Reject H0")
    print("Conclusion: There is no significant difference in mean BMI between the two groups.")
```

```
Two-Sample t-Test: BMI
----------------------
H0: Mean BMI of diabetic and non-diabetic patients is equal
H1: Mean BMI of diabetic and non-diabetic patients is different

Mean BMI - Non-Diabetic: 30.3
Mean BMI - Diabetic: 35.14
Test Statistic (t): 8.6193
P-value: 0.0
Decision: Reject H0
Conclusion: There is a significant difference in mean BMI between the two groups.
```

[8]
0s

```
df["Glucose_Category"] = np.where(
    df["Glucose"] < 100,
    "Normal",
    "High"
)

contingency_table = pd.crosstab(
    df["Glucose_Category"],
    df["Outcome"]
)

print("Contingency Table")
print("-----------------")
print(contingency_table)

chi2, p_value, dof, expected = stats.chi2_contingency(contingency_table)

print("\nChi-Square Test")
print("---------------")
print("H0: Glucose category and diabetes outcome are independent")
print("H1: Glucose category and diabetes outcome are associated")
```

[8]
0s

```
print("Chi-Square Statistic:", round(chi2, 4))
print("Degrees of Freedom:", dof)
print("P-value:", round(p_value, 4))

if p_value < 0.05:
    print("Decision: Reject H0")
    print("Conclusion: Glucose category is significantly associated with diabetes outcome.")
else:
    print("Decision: Fail to Reject H0")
    print("Conclusion: There is insufficient evidence of an association.")
```

```
Contingency Table
-----------------
Outcome             0    1
Glucose_Category
High              319  252
Normal            181   16

Chi-Square Test
---------------
H0: Glucose category and diabetes outcome are independent
H1: Glucose category and diabetes outcome are associated
```

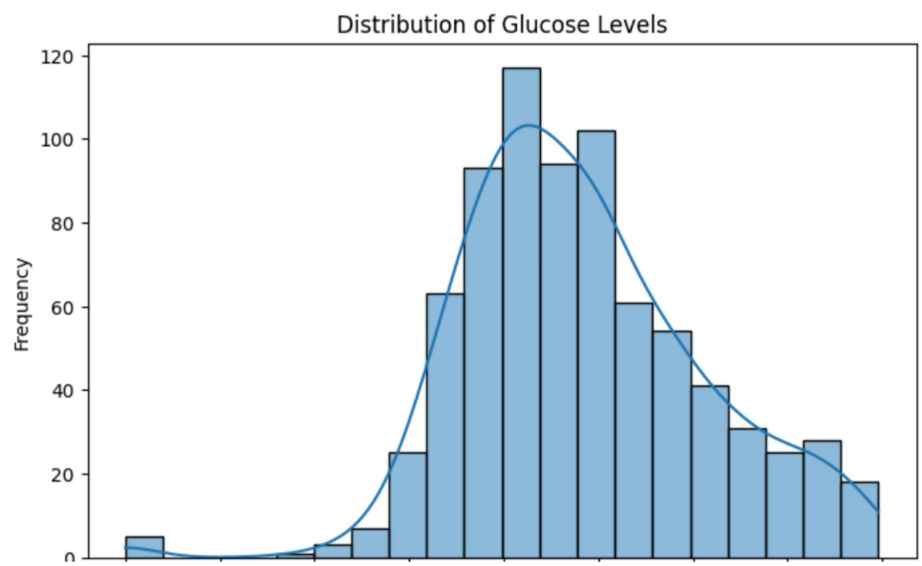
```
Chi-Square Statistic: 82.0281
Degrees of Freedom: 1
P-value: 0.0
Decision: Reject H0
Conclusion: Glucose category is significantly associated with diabetes outcome.
```

[9]
0s

```python
plt.figure(figsize=(8, 5))

sns.histplot(df["Glucose"], bins=20, kde=True)

plt.title("Distribution of Glucose Levels")
plt.xlabel("Glucose Level")
plt.ylabel("Frequency")
plt.show()
```



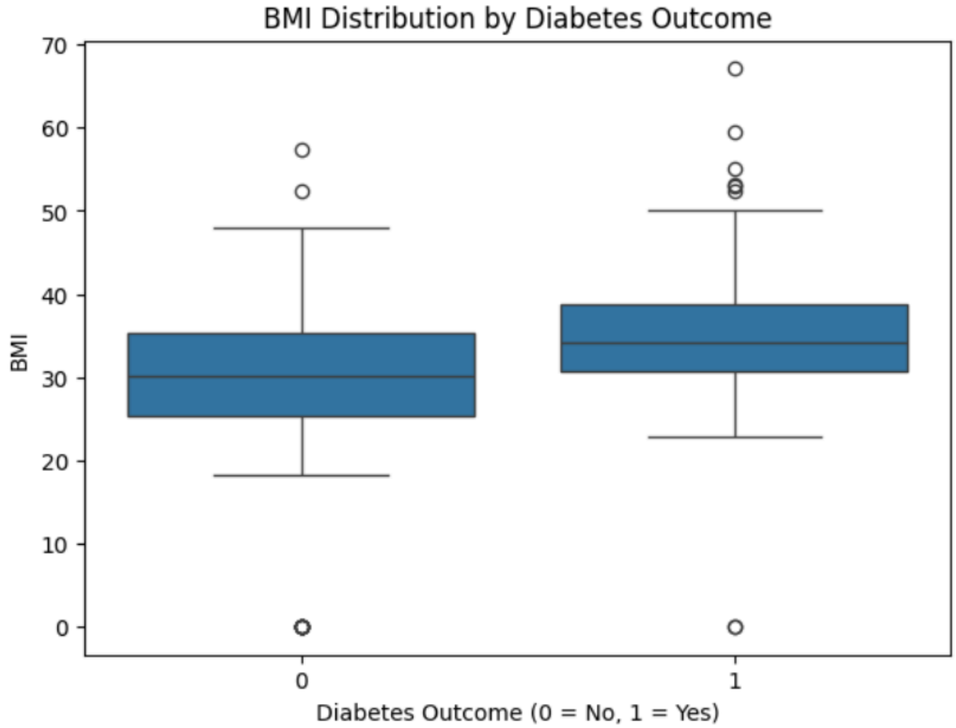
Variables Terminal

[10]
0s

```python
plt.figure(figsize=(7, 5))

sns.boxplot(x="Outcome", y="BMI", data=df)

plt.title("BMI Distribution by Diabetes Outcome")
plt.xlabel("Diabetes Outcome (0 = No, 1 = Yes)")
plt.ylabel("BMI")
plt.show()
```



[11]
✓ 0s

```
print("=" * 60)
print("STATISTICAL INFERENCE - FINAL RESULTS")
print("=" * 60)

print("\n1. POINT ESTIMATE")
print("Estimated Population Mean Glucose =", round(sample_mean, 2), "mg/dL")

print("\n2. 95% CONFIDENCE INTERVAL")
print(
    f"95% CI = ({lower_bound:.2f}, {upper_bound:.2f}) mg/dL"
)

print("\n3. ONE-SAMPLE t-TEST")
print("Test: Mean Glucose = 120 mg/dL")
print("t-statistic =", round(t_statistic, 4))
print("p-value =", round(p_value, 4))

print("\n4. DECISION")
if p_value < 0.05:
    print("Reject H0")
else:
    print("Fail to Reject H0")

print("\nSignificance Level (α) = 0.05")
```

```
print("\n" + "=" * 60)
print("EXPERIMENT COMPLETED SUCCESSFULLY")
print("=" * 60)
```

```
============================================================
STATISTICAL INFERENCE - FINAL RESULTS
============================================================

1. POINT ESTIMATE
Estimated Population Mean Glucose = 120.89 mg/dL

2. 95% CONFIDENCE INTERVAL
95% CI = (118.63, 123.16) mg/dL

3. ONE-SAMPLE t-TEST
Test: Mean Glucose = 120 mg/dL
t-statistic = 8.6193
p-value = 0.0

4. DECISION
Reject H0

Significance Level (α) = 0.05

============================================================
EXPERIMENT COMPLETED SUCCESSFULLY
============================================================
```

Conclusion

Statistical inference, estimation, and hypothesis testing techniques were applied to the Pima Indians Diabetes Dataset. Sample statistics were used to estimate population characteristics, confidence intervals were constructed, and hypothesis tests were performed to determine whether sufficient statistical evidence existed to support specific claims.

Questions to be Answered by Students

1. What population parameter was estimated in the experiment, and what was its estimated value?

ANS: The population parameter estimated was the population mean glucose level of the patients.

The sample mean was used as the point estimate:

x̄=120.89 mg/dL\bar{x} = 120.89\text{ mg/dL}

The estimated population mean glucose level is **120.89 mg/dL**.

2. What confidence interval was obtained, and how should it be interpreted?

ANS: A 95% confidence interval was constructed for the population mean glucose level.

95% CI=(118.62, 123.16) mg/dL95\%\,CI = (118.62,\ 123.16)\text{ mg/dL}

Interpretation: We are 95% confident that the true population mean glucose level lies between 118.62 mg/dL and 123.16 mg/dL.

3. What hypotheses were formulated for the selected statistical test?

ANS: For the one-sample t-test, we tested whether the population mean glucose level differs from 120 mg/dL.

- **Null Hypothesis (H_0):** The population mean glucose level is equal to 120 mg/dL.

H0:μ=120H_0:\mu=120

- **Alternative Hypothesis (H_1):** The population mean glucose level is different from 120 mg/dL.

H1:μ≠120H_1:\mu\neq120

The significance level was:

α=0.05\alpha=0.05

4. Based on the p-value, what statistical decision was made?

ANS: The one-sample t-test gave approximately:

p-value=0.4412p\text{-value}=0.4412

Since:

0.4412>0.050.4412 > 0.05

we fail to reject the null hypothesis (H_0).

Conclusion: There is insufficient statistical evidence to conclude that the population mean glucose level is significantly different from 120 mg/dL.